Formula Sheet

CBSE Class 12 Physics • NCERT Part I, Chapters 1–8

Blue boxes are NCERT (board) formulas, with SI units in [brackets]. Purple **JEE extra** boxes go beyond the NCERT book. Use with the separate revision notes.

Constants

e	1.6×10^{-19} C	m_e	9.1×10^{-31} kg
ε_0	8.854×10^{-12} C^2 N^{-1} m^{-2}	$k = \frac{1}{4\pi\varepsilon_0}$	9×10^9 N m^2 C^{-2}
μ_0	$4\pi \times 10^{-7}$ T m A^{-1}	$\frac{\mu_0}{4\pi}$	10^{-7} T m A^{-1}
c	3×10^8 m s^{-1}	h	6.63×10^{-34} J s
1 eV	1.6×10^{-19} J	1 G	10^{-4} T

Dimensions of key quantities

Charge	[AT]	E	[MLT^{-3}A^{-1}]	V	[ML^{2}T^{-3}A^{-1}]
C	[M^{-1}L^{-2}T^{4}A^2]	R	[ML^{2}T^{-3}A^{-2}]	ρ	[ML^{3}T^{-3}A^{-2}]
B	[MT^{-2}A^{-1}]	L, M	[ML^{2}T^{-2}A^{-2}]	ε_0	[M^{-1}L^{-3}T^{4}A^2]

Useful checks: RC, L/R and $\sqrt{LC}$ have dimensions of time; $\dfrac{1}{\sqrt{\mu_0\varepsilon_0}}$ is a speed.

Chapter 1 Electric Charges and Fields

Charge, force, field

$q = ne$ [C] $F = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1 q_2}{r^2}$ [N] $\vec{F} = q\vec{E}$ $E = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{r^2}$ [N C^{-1} = V m^{-1}]

In a medium: $F = \dfrac{F_0}{K}$ ($K = \varepsilon_r$).

Dipole

$p = q \times 2a$ [C m], from $-q$ to $+q$. Axial: $E = \dfrac{2p}{4\pi\varepsilon_0 r^3}$ Equatorial: $E = \dfrac{p}{4\pi\varepsilon_0 r^3}$ (opposite to $\vec{p}$)

Torque in uniform $\vec{E}$: $\vec{\tau} = \vec{p} \times \vec{E}$, $\tau = pE\sin\theta$ [N m]; net force = 0.

Flux and Gauss's law

$\Phi_E = \vec{E} \cdot \vec{S} = ES\cos\theta$ [N m^2 C^{-1}] $\oint \vec{E} \cdot d\vec{S} = \dfrac{q_{\text{enc}}}{\varepsilon_0}$

Infinite line (λ)	$E = \dfrac{\lambda}{2\pi\varepsilon_0 r}$	radial
Infinite sheet (σ)	$E = \dfrac{\sigma}{2\varepsilon_0}$	independent of r
Two oppositely charged plates	$E = \dfrac{\sigma}{\varepsilon_0}$	between, 0 outside
Thin shell (q, R)	$E = \dfrac{q}{4\pi\varepsilon_0 r^2}$	$(r \geq R)$; 0 $(r < R)$

Densities: $\lambda = \frac{q}{l}$ [C m^{-1}], $\sigma = \frac{q}{A}$ [C m^{-2}], $\rho = \frac{q}{V}$ [C m^{-3}]

JEE extra

Dipole at angle θ: $E = \dfrac{p\sqrt{1+3\cos^2\theta}}{4\pi\varepsilon_0 r^3}$. Solid sphere: $E = \dfrac{kQr}{R^3}$ inside. Charge at a cube corner: flux through cube $\dfrac{q}{8\varepsilon_0}$. Dipole in non-uniform field: $F = p\dfrac{dE}{dx}$.

Chapter 2 Electrostatic Potential and Capacitance

Potential and energy

$V = \dfrac{W}{q}$ [V] Point charge: $V = \dfrac{q}{4\pi\varepsilon_0 r}$ Dipole: $V = \dfrac{p\cos\theta}{4\pi\varepsilon_0 r^2}$ (0 on equatorial line)

$E = -\dfrac{dV}{dr}$ Two charges: $U = \dfrac{q_1 q_2}{4\pi\varepsilon_0 r_{12}}$ [J] Dipole in field: $U = -\vec{p}\cdot\vec{E} = -pE\cos\theta$

Charge in external field: $U = qV(\vec{r})$. Work moving q between points: $W = q(V_B - V_A)$.

Capacitors

$C = \dfrac{Q}{V}$ [F] Parallel plate: $C = \dfrac{\varepsilon_0 A}{d}$; with dielectric $C = \dfrac{K\varepsilon_0 A}{d}$

Series: $\dfrac{1}{C} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \ldots$ (same Q) Parallel: $C = C_1 + C_2 + \ldots$ (same V)

Energy: $U = \dfrac{Q^2}{2C} = \dfrac{1}{2}CV^2 = \dfrac{1}{2}QV$ [J] Energy density: $u = \dfrac{1}{2}\varepsilon_0 E^2$ [J m^{-3}]

Dielectric: battery connected $\Rightarrow$ V same, $Q \to KQ$, $U \to KU$; disconnected $\Rightarrow$ Q same, $V \to V/K$, $U \to U/K$.

JEE extra

Slab of thickness t: $C = \dfrac{\varepsilon_0 A}{d - t + t/K}$ (metal slab: $K \to \infty$). Isolated sphere: $C = 4\pi\varepsilon_0 R$. Spherical capacitor: $C = 4\pi\varepsilon_0 \dfrac{ab}{b-a}$.

Sharing: $V = \dfrac{C_1V_1 + C_2V_2}{C_1 + C_2}$, loss $= \dfrac{C_1C_2(V_1 - V_2)^2}{2(C_1 + C_2)}$. Force between plates: $F = \dfrac{Q^2}{2\varepsilon_0 A}$. RC charging: $q = CV(1 - e^{-t/RC})$.

Chapter 3 Current Electricity

Current and conduction

$I = \dfrac{dq}{dt}$ [A] $I = neAv_d$ $v_d = \dfrac{eE\tau}{m}$ [m s^{-1}] mobility $\mu = \dfrac{v_d}{E} = \dfrac{e\tau}{m}$ [m^2 V^{-1} s^{-1}]

$j = \dfrac{I}{A} = \sigma E$ $\rho = \dfrac{m}{ne^2\tau}$ [Ω m] $\sigma = \dfrac{1}{\rho}$ [S m^{-1}] $R = \dfrac{\rho l}{A}$ [Ω] $V = IR$

Temperature: $\rho_T = \rho_0[1 + \alpha(T - T_0)]$, α [°C^{-1} or K^{-1}]

Power, cells, networks

$P = VI = I^2R = \dfrac{V^2}{R}$ [W] $V = \varepsilon - Ir$ (discharging), $V = \varepsilon + Ir$ (charging) $I = \dfrac{\varepsilon}{R + r}$

Resistors: series $R = R_1 + R_2$; parallel $\dfrac{1}{R} = \dfrac{1}{R_1} + \dfrac{1}{R_2}$.

Cells in series: $\varepsilon = \varepsilon_1 + \varepsilon_2$, $r = r_1 + r_2$. Parallel: $\dfrac{\varepsilon}{r} = \dfrac{\varepsilon_1}{r_1} + \dfrac{\varepsilon_2}{r_2}$, $\dfrac{1}{r} = \dfrac{1}{r_1} + \dfrac{1}{r_2}$.

Kirchhoff: $\sum I = 0$ at a junction; $\sum \varepsilon = \sum IR$ round a loop. Wheatstone balance: $\dfrac{P}{Q} = \dfrac{R}{S}$.

JEE extra

Max power: $R = r$, $P_{max} = \dfrac{\varepsilon^2}{4r}$. Wire stretched n times: $R' = n^2R$. m rows of n cells: $I = \dfrac{n\varepsilon}{R + nr/m}$. Metre bridge: $\dfrac{R}{S} = \dfrac{l}{100 - l}$. Heat: $H = I^2Rt$.

Chapter 4 Moving Charges and Magnetism

Forces

Lorentz: $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$; magnetic part $F = qvB\sin\theta$ [N] Conductor: $\vec{F} = I\vec{l} \times \vec{B}$

Circular motion: $r = \dfrac{mv}{qB}$, $T = \dfrac{2\pi m}{qB}$, $\nu = \dfrac{qB}{2\pi m}$ [Hz], $\omega = \dfrac{qB}{m}$; helix pitch $= v\cos\theta \cdot T$

Parallel wires: $\dfrac{F}{L} = \dfrac{\mu_0 I_1 I_2}{2\pi d}$ [N m^{-1}] (like currents attract)

Magnetic fields of currents

Biot–Savart: $d\vec{B} = \dfrac{\mu_0}{4\pi}\dfrac{I\, d\vec{l} \times \vec{r}}{r^3}$ Ampere: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$

Long straight wire $B = \dfrac{\mu_0 I}{2\pi r}$ Centre of coil (N turns) $B = \dfrac{\mu_0 NI}{2R}$

Axis of loop $B = \dfrac{\mu_0 IR^2}{2(x^2 + R^2)^{3/2}}$ Long solenoid $B = \mu_0 nI$ (n per metre)

Torque and galvanometer

$\vec{m} = NI\vec{A}$ [A m^2] $\vec{\tau} = \vec{m} \times \vec{B}$, $\tau = NIAB\sin\theta$ (θ from the normal)

MCG: $\phi = \dfrac{NAB}{k} I$; current sensitivity $\dfrac{NAB}{k}$; voltage sensitivity $\dfrac{NAB}{kR}$

Ammeter: shunt $S = \dfrac{I_g G}{I - I_g}$ Voltmeter: series $R = \dfrac{V}{I_g} - G$

JEE extra

Finite wire: $B = \dfrac{\mu_0 I}{4\pi d}(\sin\alpha_1 + \sin\alpha_2)$. Arc of angle θ: $B = \dfrac{\mu_0 I\theta}{4\pi R}$. Toroid: $B = \dfrac{\mu_0 NI}{2\pi r}$.

Orbiting electron: $m = \dfrac{evr}{2}$, $\dfrac{m}{L} = \dfrac{e}{2m_e}$; Bohr magneton 9.27×10^{-24} A m^2. Cyclotron frequency $\nu_c = \dfrac{qB}{2\pi m}$.

Chapter 5 Magnetism and Matter

Magnetic dipole

Bar magnet / solenoid: $m = NIA$ [J T^{-1} = A m^2]

Axial: $B = \dfrac{\mu_0}{4\pi}\dfrac{2m}{r^3}$ Equatorial: $B = \dfrac{\mu_0}{4\pi}\dfrac{m}{r^3}$ (opposite to $\vec{m}$)

$\vec{\tau} = \vec{m} \times \vec{B}$ $U = -\vec{m}\cdot\vec{B} = -mB\cos\theta$ $W_{\theta_1\to\theta_2} = mB(\cos\theta_1 - \cos\theta_2)$

Gauss's law for magnetism: $\oint \vec{B}\cdot d\vec{S} = 0$ (no monopoles).

Magnetic properties of materials

$M = \dfrac{m_{net}}{V}$ [A m^{-1}] $\vec{B} = \mu_0(\vec{H} + \vec{M})$ $M = \chi H$ $\mu_r = 1 + \chi$ $\mu = \mu_0\mu_r$

	Diamagnetic	**Paramagnetic**	**Ferromagnetic**
χ	small, negative ($-1 \le \chi < 0$)	small, positive	very large, positive
μ_r	$0 \le \mu_r < 1$	slightly > 1	$\gg 1$
In non-uniform B	moves to weaker field	to stronger field	strongly to stronger field
Examples	Bi, Cu, Pb, Si, H_2O, N_2	Al, Na, Ca, O_2	Fe, Co, Ni, Gd

Superconductor: $\chi = -1$, $\mu_r = 0$ (Meissner effect).

JEE extra

Curie's law: $\chi = \dfrac{C\mu_0}{T}$. Curie–Weiss: $\chi = \dfrac{C}{T - T_c}$. Oscillating magnet: $T = 2\pi\sqrt{\dfrac{I}{mB}}$. Earth's field: $B_H = B_E\cos\delta$, $\tan\delta = B_V/B_H$.

Chapter 6 Electromagnetic Induction

Faraday and Lenz

$\Phi_B = \vec{B}\cdot\vec{A} = BA\cos\theta$ [Wb = T m^2] $\varepsilon = -N\dfrac{d\Phi_B}{dt}$ [V] (minus sign: Lenz's law)

Motional emf: $\varepsilon = Blv$ Rotating rod: $\varepsilon = \dfrac{1}{2}B\omega l^2$ Induced current $I = \dfrac{\varepsilon}{R}$

Inductance

Self: $N\Phi = LI,\ \varepsilon = -L\dfrac{dI}{dt},\ L$ [H] Solenoid: $L = \mu_0 n^2 Al = \dfrac{\mu_0 N^2 A}{l}$ ($\times\mu_r$ with core)

Mutual: $N_1\Phi_1 = MI_2,\ \varepsilon_1 = -M\dfrac{dI_2}{dt}$ Coaxial solenoids: $M = \mu_0 n_1 n_2 \pi r_1^2 l$

Energy: $U = \dfrac{1}{2}LI^2$ [J] AC generator: $\varepsilon = NBA\omega\sin\omega t,\ \varepsilon_0 = NBA\omega$

JEE extra

Charge through coil: $q = \dfrac{N\Delta\Phi}{R}$. Rod with resistance: retarding force $\dfrac{B^2l^2v}{R}$, terminal speed under gravity $v = \dfrac{mgR}{B^2l^2}$. $M = k\sqrt{L_1L_2}$. $u_B = \dfrac{B^2}{2\mu_0}$. LR growth: $I = I_0(1 - e^{-Rt/L})$.

Chapter 7 Alternating Current

rms values and reactances

$v = v_m\sin\omega t,\ \omega = 2\pi\nu$ $I = \dfrac{i_m}{\sqrt{2}},\ V = \dfrac{v_m}{\sqrt{2}}$

Element	Opposition [Ω]	Phase of i	Average power
R	R	in phase	I^2R
L	$X_L = \omega L$	lags v by $\frac{\pi}{2}$	0
C	$X_C = \dfrac{1}{\omega C}$	leads v by $\frac{\pi}{2}$	0

Series LCR, resonance, power, transformer

$Z = \sqrt{R^2 + (X_L - X_C)^2}$ $\tan\phi = \dfrac{X_L - X_C}{R}$ $i_m = \dfrac{v_m}{Z}$

Resonance: $\omega_0 = \dfrac{1}{\sqrt{LC}},\ \nu_0 = \dfrac{1}{2\pi\sqrt{LC}},\ Z = R$ $Q = \dfrac{\omega_0 L}{R} = \dfrac{1}{\omega_0 CR} = \dfrac{1}{R}\sqrt{\dfrac{L}{C}}$

$P = VI\cos\phi$ [W] power factor $\cos\phi = \dfrac{R}{Z}$ wattless current $= I\sin\phi$

Transformer: $\dfrac{V_s}{V_p} = \dfrac{N_s}{N_p} = \dfrac{I_p}{I_s}$ (ideal) efficiency $\eta = \dfrac{P_{out}}{P_{in}}$

JEE extra

Bandwidth $\Delta\omega = \dfrac{R}{L}$, $Q = \dfrac{\omega_0}{\Delta\omega}$. Average over half cycle: $\bar{i} = \dfrac{2i_m}{\pi}$. LC oscillation: $q = q_0\cos\omega_0 t$, energy $\dfrac{q_0^2}{2C}$. Parallel LC at resonance: $Z \to \infty$.

Chapter 8 Electromagnetic Waves

Displacement current and EM waves

$i_d = \varepsilon_0 \dfrac{d\Phi_E}{dt}$ [A] Ampere–Maxwell: $\oint \vec{B}\cdot d\vec{l} = \mu_0(i_c + i_d)$

$c = \dfrac{1}{\sqrt{\mu_0\varepsilon_0}} = 3\times10^8$ m s^{-1} medium: $v = \dfrac{1}{\sqrt{\mu\varepsilon}}$ $c = \nu\lambda$ $E_0 = cB_0$

$E_x = E_0\sin(kz-\omega t)$, $B_y = B_0\sin(kz-\omega t)$, $k = \dfrac{2\pi}{\lambda}$ [rad m^{-1}], $\omega = ck$ Photon: $E = h\nu = \dfrac{hc}{\lambda}$

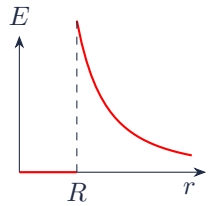
Electromagnetic spectrum (increasing wavelength)

Gamma	X-rays	UV	Visible	IR	Microwave	Radio
$< 10^{-3}$ nm	1 nm–10^{-3} nm	400–1 nm	700–400 nm	1 mm–700 nm	0.1 m–1 mm	> 0.1 m

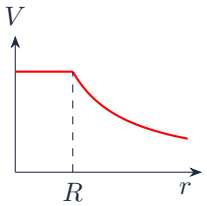
JEE extra

Energy density $u = \dfrac{1}{2}\varepsilon_0 E_0^2$ (average, equal shares E and B). Intensity $I = \dfrac{1}{2}\varepsilon_0 E_0^2 c$. Radiation pressure $\dfrac{I}{c}$ (absorbed), $\dfrac{2I}{c}$ (reflected). $E(\text{eV}) \approx \dfrac{1240}{\lambda(\text{nm})}$.

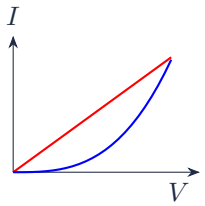
Graph shapes to remember



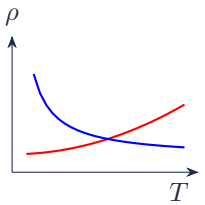
E vs r: shell



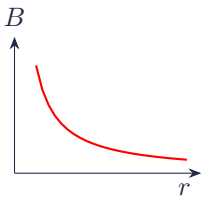
V vs r: shell



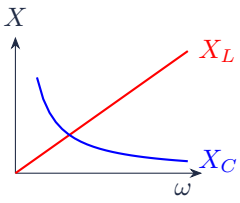
ohmic (red), non-ohmic



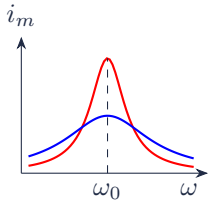
metal (red), semiconductor



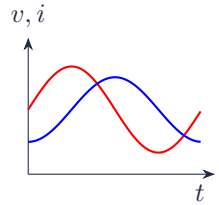
$B \propto 1/r$: straight wire



reactance vs ω



resonance: small R (red)



inductor: i (blue) lags v